Topos Theory: Categorical Preliminaries, Internal Logic, and the Geometry of Sets

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§1.1 Why topoi enter the story

A topos is one of the places where category theory, logic, geometry, sheaf theory, and foundations meet. I want to regard it not merely as an abstract generalization of set theory, but as a way of making precise the idea that "sets may vary over a space" and that logic may depend on the ambient universe in which it is interpreted.

The route of this chapter is therefore:

$$\begin{gathered}\text{categories} \to \text{limits and universal properties}\\ \to \text{cartesian closure} \to \text{subobject classifiers}\\ \to \text{elementary topoi} \to \text{sheaf topoi and internal logic.}\end{gathered}$$

§1.2 Categories

Definition 1.2.1. A category $\mathcal{C}$ consists of:

(i) a class of objects $\mathrm{Ob}(\mathcal{C})$;

(ii) for each pair of objects X, Y, a collection $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ of morphisms from X to Y;

(iii) a composition operation

$$\mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X, Z),$$

written $(g, f) \mapsto g \circ f$;

(iv) for each object X, an identity morphism $\mathrm{id}_X : X \to X$.

These data satisfy associativity and identity laws:

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad f \circ \mathrm{id}_X = f, \quad \mathrm{id}_Y \circ f = f.$$

Example 1.2.2

The category **Set** has sets as objects and functions as morphisms. The category **Grp** has groups and group homomorphisms. The category **Top** has topological spaces and continuous maps.

Remark 1.2.3 — The categorical viewpoint is not that objects become unimportant, but that objects are understood through their maps. This is exactly the shift that later makes generalized set theories possible.

§1.3 Universal properties and limits

The most important categorical habit is to define objects not by their internal elements, but by how maps into or out of them behave.

Definition 1.3.1. A terminal object in $\mathcal{C}$ is an object 1 such that for every object X, there is a unique morphism

$$X \to 1.$$

Dually, an initial object is an object 0 such that for every X, there is a unique morphism

$$0 \to X.$$

In **Set**, a singleton is terminal and the empty set is initial.

Definition 1.3.2. A product of X and Y is an object $X \times Y$, equipped with projections

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y,$$

such that for every object Z and every pair of maps $f : Z \to X$, $g : Z \to Y$, there exists a unique map $\langle f, g \rangle : Z \to X \times Y$ making the diagram commute:

$$\begin{tikzcd}
 & Z \arrow[d, dashed, "\langle f,g \rangle"] \arrow[ddl, bend right, "f"'] \arrow[ddr, bend left, "g"] & \\
 & X \times Y \arrow[dl, "\pi_X"] \arrow[dr, "\pi_Y"'] & \\
X & & Y.
\end{tikzcd}$$

Definition 1.3.3. An equalizer of parallel morphisms $f, g : X \to Y$ is a morphism $e : E \to X$ such that

$$f \circ e = g \circ e,$$

and such that any other map $h : Z \to X$ with $f \circ h = g \circ h$ factors uniquely through e:

$$\begin{tikzcd}
Z \arrow[r, dashed, "!"] \arrow[rr, bend left, "h"] & E \arrow[r, "e"] & X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y
\end{tikzcd}$$

Definition 1.3.4. A pullback of $f : X \to Z$ and $g : Y \to Z$ is an object $X \times_Z Y$ fitting into a cartesian square

$$\begin{tikzcd}
X \times_Z Y \arrow[r] \arrow[d] & Y \arrow[d, "g"] \\
X \arrow[r, "f"'] & Z
\end{tikzcd}$$

with the universal property that any object mapping compatibly to X and Y factors uniquely through the pullback.

Remark 1.3.5 — Products, equalizers, and pullbacks are not isolated gadgets. They are instances of limits. A limit is a universal cone over a diagram; it is the categorical way to say that an object satisfies all the compatibility conditions encoded by the diagram.

§1.4 Exponentials and cartesian closure

In set theory, the set of functions from X to Y is denoted Y^X. Categorically, this is characterized by an adjunction.

Definition 1.4.1. A category with finite products is cartesian closed if for every pair of objects X, Y, there exists an exponential object Y^X and a natural bijection

$$\mathrm{Hom}_{\mathcal{C}}(Z, Y^X) \cong \mathrm{Hom}_{\mathcal{C}}(Z \times X, Y).$$

The corresponding evaluation map is

$$\mathrm{ev} : Y^X \times X \to Y.$$

A morphism $Z \to Y^X$ is the same thing as a morphism $Z \times X \to Y$. This is currying in categorical form:

$$\begin{tikzcd} Z \times X \arrow[rr, "\tilde{f}"] \arrow[dr, "\hat{f} \times \mathrm{id}_X"'] & & Y \\ & Y^X \times X \arrow[ur, "\mathrm{ev}"'] & \end{tikzcd}$$

Example 1.4.2

In **Set**, Y^X is the ordinary set of functions $X \to Y$. The bijection

$$\mathrm{Hom}_{\mathbf{Set}}(Z, Y^X) \cong \mathrm{Hom}_{\mathbf{Set}}(Z \times X, Y)$$

says that a Z-indexed family of functions $X \to Y$ is the same thing as a function of two variables $Z \times X \to Y$.

§1.5 Subobjects and the subobject classifier

A subset $A \subseteq X$ may be represented by its characteristic function

$$\chi_A : X \to \{0, 1\}.$$

The categorical version replaces $\{0, 1\}$ by an object Ω, called the subobject classifier.

Definition 1.5.1. A subobject classifier in a category with a terminal object is an object Ω, together with a map

$$\mathsf{true} : 1 \to \Omega,$$

such that for every monomorphism $m : A \hookrightarrow X$, there is a unique characteristic morphism $\chi_A : X \to \Omega$ making the square a pullback:

$$\begin{tikzcd} A \arrow[r] \arrow[d, "m"'] & 1 \arrow[d, "\mathsf{true}"] \\ X \arrow[r, "\chi_A"'] & \Omega. \end{tikzcd}$$

In **Set**, $\Omega = \{0, 1\}$, true picks 1, and χ_A is the ordinary characteristic function.

Remark 1.5.2 — This is one of the decisive steps toward topos theory. A topos has an internal object of truth values, so logic becomes an internal structure of the category rather than an external metalanguage imposed from outside.

§1.6 Elementary topoi

Definition 1.6.1. An elementary topos is a category $\mathcal{E}$ satisfying:

(i) it has finite limits;

(ii) it is cartesian closed;

(iii) it has a subobject classifier.

This definition is strikingly short. It says that a category behaves enough like **Set** to support finite constructions, function spaces, and a logic of predicates.

Example 1.6.2

The category **Set** is an elementary topos. More generally, for any small category $\mathcal{C}$, the presheaf category

$$\widehat{\mathcal{C}} = \mathbf{Set}^{\mathcal{C}^{op}}$$

is an elementary topos.

Example 1.6.3

If X is a topological space, the category $\mathrm{Sh}(X)$ of sheaves on X is a topos. It behaves like a category of sets varying continuously over X.

§1.7 A toy presheaf topos

The smallest example that is not simply **Set** is already useful. Let

$$\mathbf{2} = (0 \to 1)$$

be the category associated to the two-element poset. Consider the presheaf category

$$\widehat{\mathbf{2}} = \mathbf{Set}^{\mathbf{2}^{op}}.$$

Every presheaf category is a topos, so this is a genuine elementary topos.

An object $P \in \widehat{\mathbf{2}}$ is exactly the data

$$P(1) \xrightarrow{r} P(0),$$

because the arrow $0 \to 1$ in $\mathbf{2}$ becomes a restriction map $P(1) \to P(0)$ in $\mathbf{2}^{op}$. It is safest to think of this as a two-stage set: elements visible at stage 1 restrict to elements visible at stage 0. It is not automatically an equivalence relation or a quotient; it is simply a function carrying information from the higher stage to the lower stage.

The subobject classifier is made of sieves. With this convention, the sieves on 0 are

$$\Omega(0) = \{\varnothing, \{id_0\}\},$$

while the sieves on 1 are

$$\Omega(1) = \{\varnothing, \{u : 0 \to 1\}, \{id_1, u\}\}.$$

Thus the three-valued part occurs at stage 1, and the restriction map

$$\Omega(1) \longrightarrow \Omega(0)$$

is

$$\varnothing \mapsto \varnothing, \qquad \{u\} \mapsto \{id_0\}, \qquad \{id_1, u\} \mapsto \{id_0\}.$$

The element $\{u\}$ is the interesting third truth value: not true at stage 1 itself, but true after restricting along $0 \to 1$.

Here is a characteristic map in this toy universe. Let

$$P(1) = \{a, b\}, \qquad P(0) = \{*\},$$

with both a and b restricting to $*$. Define a subpresheaf $Q \hookrightarrow P$ by

$$Q(1) = \{a\}, \qquad Q(0) = \{*\}.$$

The characteristic morphism $\chi_Q : P \to \Omega$ sends

$$\chi_Q(1)(a) = \{id_1, u\}, \qquad \chi_Q(1)(b) = \{u\},$$

and

$$\chi_Q(0)(*) = \{id_0\}.$$

So a is fully true, while b is not true at stage 1 but becomes true after restriction to stage 0. This is the point at which internal logic stops being two-valued. In $\mathbf{Set}^{\mathbf{2}^{op}}$, a proposition may have a stage-dependent truth value; truth is not merely "yes" or "no", but is tracked through the restriction structure.

§1.8 Grothendieck topoi and sheaves

A Grothendieck topos is, roughly, a category equivalent to sheaves on a site. A site is a category equipped with a notion of covering. Thus Grothendieck topoi generalize sheaves on topological spaces.

> **Remark 1.8.1 —** This is the geometric origin of topos theory. Instead of studying a space only through its points, one studies its sheaves. In many situations, the sheaf category remembers more robust information than the point set itself.

In algebraic geometry, this becomes essential. The étale topos of a scheme, for example, packages arithmetic and geometric information that is invisible from the underlying topological space alone.

§1.9 Internal logic

Every elementary topos carries an internal higher-order intuitionistic logic. Subobjects of an object X behave like predicates on X, and the subobject classifier Ω behaves like the object of truth values.

> **Remark 1.9.1 —** The internal logic of a general topos is usually intuitionistic, not necessarily classical. The law of excluded middle
>
> $$P \vee \neg P$$
>
> need not hold internally. Thus moving from $\mathbf{Set}$ to a general topos also changes the logic that is naturally available.

For a sheaf topos, this has a geometric meaning: truth may be local. A statement may hold on an open cover without having a single global witness. This is exactly the kind of phenomenon that sheaf theory is designed to express.

§1.10 Power objects

In set theory, $\mathcal{P}(X)$ is the set of subsets of X. In a topos, the analogue is the power object

$$\Omega^X.$$

Indeed, by cartesian closure and the subobject classifier,

$$\mathrm{Hom}_{\mathcal{E}}(B, \Omega^X) \cong \mathrm{Hom}_{\mathcal{E}}(B \times X, \Omega),$$

which classifies subobjects of $B \times X$. Thus Ω^X parametrizes subobjects of X, just as 2^X parametrizes subsets of X in ordinary set theory.

§1.11 Topos as generalized universe of sets

A topos can be read in two complementary ways:

(i) as a generalized universe of sets, where one can do mathematics internally;

(ii) as a generalized space, where sheaves and local data become fundamental.

This double reading is the source of its power. In one direction, topos theory generalizes set-theoretic foundations. In the other direction, it turns geometry into logic.

§1.12 A high-level comparison

Setting	Truth values	Subsets/predicates
Set	$\{0, 1\}$	characteristic functions $X \to 2$
Boolean topos	Boolean algebra object	classical internal logic
General topos	Ω	subobjects classified by $X \to \Omega$
Sheaf topos	local truth values	predicates varying over a space

Remark 1.12.1 — From a higher viewpoint, a topos is not merely a category satisfying three axioms. It is a world in which mathematics can be interpreted. The shift from sets to topoi is analogous to the shift from Euclidean geometry to geometry relative to a transformation group: the ambient universe becomes variable.

§1.13 Further directions

Several directions naturally follow:

(i) Lawvere–Tierney topologies, which internalize sheafification;

(ii) geometric morphisms, the correct maps between topoi;

(iii) classifying topoi, which classify models of geometric theories;

(iv) Grothendieck topoi in algebraic geometry;

(v) elementary topoi as foundations for constructive mathematics.

Question

The most important question for me is not only what a topos is, but what it means to do mathematics inside one. Once truth, subsets, and function spaces become internal categorical objects, foundations stop being a fixed background and become part of the geometry.

§1.14 Subobjects and characteristic maps

In **Set**, a subset $A \subseteq X$ is classified by its characteristic function

$$\chi_A : X \to \{0, 1\}.$$

A subobject classifier generalizes this. In a topos, there is an object Ω and a map

$$\mathsf{true} : 1 \to \Omega$$

such that every subobject $A \hookrightarrow X$ is obtained as a pullback of true along a unique characteristic map $\chi_A : X \to \Omega$.

This is the categorical replacement for membership predicates. A subobject is not merely a subset; it is something classified by a truth-value object internal to the topos.

§1.15 Why logic becomes intuitionistic

In a general topos, the object Ω of truth values need not behave like the two-element Boolean algebra. As a result, the internal logic is generally intuitionistic. In particular, the law of excluded middle

$$P \vee \neg P$$

may fail internally.

This is not a defect. It means that truth can vary with geometric or contextual information. In a sheaf topos, a proposition may be true locally on some open sets without being globally decidable.

§1.16 Sheaves as variable sets

A sheaf on a topological space assigns data to each open set, with restriction maps and gluing conditions. One can think of a sheaf as a set varying over the space. The topos of sheaves on X therefore behaves like a universe of variable sets over X.

This is the geometric intuition behind topoi. They are not arbitrary abstract categories; they generalize the category of sets in a way that remembers variation, locality, and gluing.